

Chapter 76

Controlling Image Processing and Resolution

This chapter covers the overall image-processing pipeline. It discusses color bit-depth and how to control the output resolution in a resolution-independent environment.

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Fusion's Place in the DaVinci Resolve Image-Processing Pipeline

When working in a single unified environment like DaVinci Resolve, it is important to understand the order of operations among the pages. DaVinci Resolve exposes some of this via the order of the page buttons at the bottom of the screen, with the Media, Cut, and Edit page at the beginning of the chain and the Color, Fairlight, and Deliver page at the end. However, this isn't the whole story, especially when it comes to the Fusion page. The following sections describe where the Fusion page fits in the image-processing chain of DaVinci Resolve.

Source Media into the Fusion Page

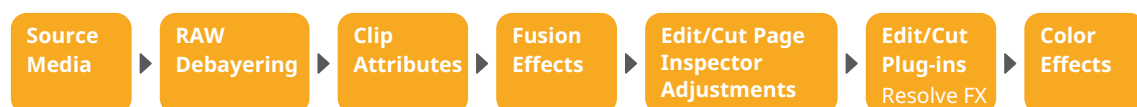
For ordinary, single clips coming in from the Edit or Cut page, the MediaIn node in the Fusion page represents the source media, as modified by the Clip Attributes window. Although you select the clip from the Edit or Cut page Timeline, in the Fusion page, the clip is accessed from the Media Pool.

TIP: The decoding or debayering of RAW files occurs prior to all other operations, and as such, any RAW adjustments will be displayed correctly in the Fusion page.

This means you have access to the entire source clip in the Fusion page, but the render range is set to match the duration of the clip in the Timeline. You also use the full resolution of the source clip, even if the Timeline is set to a lower resolution. However, none of the Edit or Cut page Inspector adjustments carry over into the Fusion page, with the exception of the Lens Correction adjustment.

When you make Zoom, Position, Crop, or Stabilization changes in the Edit or Cut page, they are not visible in the Fusion page. The same applies to any Resolve FX or Open FX third-party plugins. If you add these items to a clip in the Edit or Cut page, and then you open the Fusion page, you won't see them taking effect. All Edit and Cut page timeline effects and Inspector adjustments, with the exception of the Lens Correction adjustment, are computed after the Fusion page but before the Color page. If you open the Color page, you'll see the Edit and Cut page transforms and plugins applied to that clip, effectively as an operation before the grading adjustments and effects you apply in the Color page Node Editor.

With this in mind, the order of effects processing in the different pages of DaVinci Resolve can be described as follows:



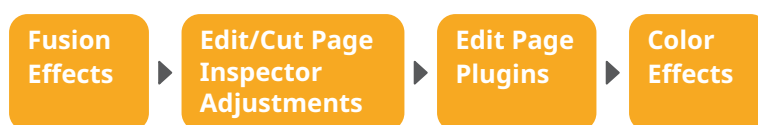
TIP: Retiming applied to the clip in the Edit page Timeline is also not carried over into the Fusion page.

Forcing Effects into the Fusion Page

There is a way you can force clips with Edit page Inspector adjustments, plugins, retiming, and Color page grades into the Fusion page, and that is to turn that clip into a compound clip or a Fusion clip. When Edit page effects and Color page grading are embedded within compound or Fusion clips, MediaIn nodes corresponding to compound clips route the effected clip into the Fusion page. However, bringing a compound clip or Fusion clip into the Fusion page changes the resolution of the source clip to match the Timeline resolution. For more information, see the section “Sizing Between DaVinci Resolve Pages” in this chapter.

Output from the Fusion Page to the Color Page

The composition output from the Fusion page’s MediaOut node are passed on via the Color page’s source input, with the sole exception that if you’ve added plugins to that clip in the Edit or Cut page, then the handoff from the Fusion page to the Color page is as follows:



What Viewers Show in Different DaVinci Resolve Pages

Owing to the different needs of compositing artists, editors, and colorists, the viewers show different states of the clip.

- **The Edit page source viewer:** Always shows the source media, unless you’re opening a compound clip that’s been saved in the Media Pool. If Resolve Color Management is enabled, then the Edit page source viewer shows the source media at the Timeline color space and gamma.
- **The Edit page Timeline viewer:** Shows clips with all Edit page effects, Color page grades, and Fusion page effects applied, so editors see the program within the context of all effects and grading.
- **The Fusion page viewer:** Shows Media Pool source clips at the Timeline color space and gamma, including Resolve Color Management adjustments, but no Edit page Inspector adjustments or Resolve FX effects and no Color page grades.
- **The Color page viewer:** Shows clips with all Edit page effects, Color page grades, and Fusion page effects applied.

Managing Resolution In Fusion

There is no formal resolution for a comp in Fusion.

- In the Fusion page, the default resolution for Fusion-generated images, like the Background tool, Fast Noise, and Text+, etc., is determined by the Timeline resolution.
- In Fusion Studio, the resolution for generators is set in Preferences > Frame Format, where you can select a standard resolution from the drop-down menu, or set the Width and Height manually.
- The resolution of a composition in the Fusion page is initially determined by the source resolution of the input image.

For example, if you have a Timeline set to HD resolution (1920 x 1080) and have a clip on the timeline whose source resolution is UHD (3840 x 2160), opening this clip in the Fusion page will set the composition's resolution to UHD (3840 x 2160).

All Fusion-generated images in this composition will be UHD resolution.



Fusion timeline sizing for the Fusion page

TIP: The output of the Fusion page is placed back into the Edit page Timeline based on DaVinci Resolve's Image Sizing setting. By default, DaVinci Resolve uses an image sizing setting called Scale to Fit. This means that even if the Fusion page outputs a 4K composition, it conforms to 1920 x 1080 if that is what the project or a particular Timeline is set to. Changing the image sizing setting in DaVinci Resolve's Project Settings affects how Fusion compositions are integrated into the Edit page Timeline.

Changing the Resolution of a Clip

If your comp uses a single image, you can change the pixel output resolution in several ways. Three common tools that change the pixel resolution of a clip are the Resize, Scale, and Crop nodes. A fourth node, Letterbox, is less commonly used but also changes the pixel resolution of a clip.

These four nodes are located in the Transform category of the Effects library. Resize is also located in the toolbar.

- **Crop:** Sets the output resolution of the node using a combination of X and Y size along with X and Y offset to cut the frame down to the size you want. Crop removes pixels from the image, so if you later use a Transform node and try to move the image, those pixels are not available.
- **Letterbox:** Sets the output resolution of the node by adding horizontal or vertical black edges where necessary to format the frame size and aspect ratio.
- **Resize:** Sets the output resolution of the node using absolute pixels.
- **Scale:** Sets the output resolution of the node using a relative percentage of the current input image size.

TIP: To change resolution and reposition a frame without changing the pixel resolution of a clip, use the Transform node.

Compositing with Different-Resolution Clips

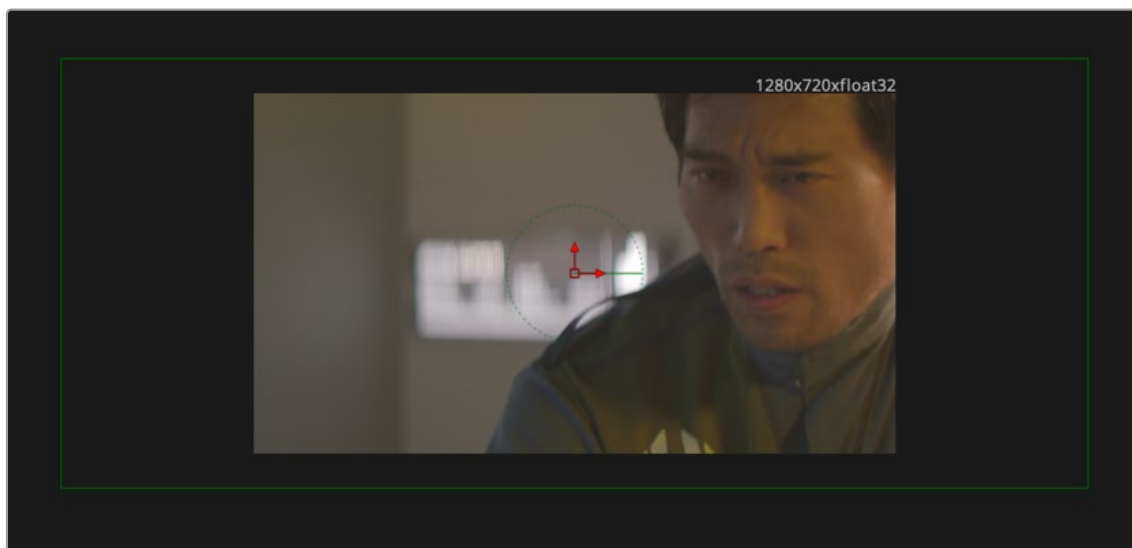
When you composite images with different resolutions using the Merge node, the image that's connected to the orange background input determines the output resolution of the Merge node.

Often, it's easiest to control the comp resolution right at the start by connecting a node with the desired output resolution you want to the orange background input on the Merge node. A Background node is often used in this situation because it consumes meager system resources.



A Background node determines the output resolution of the merge

The Background node sets the output size, and the foreground image is cropped if it is larger.



A Background node created at 1280 x 720 crops the larger foreground. However, all the pixels of the larger foreground are available for repositioning.

Sizing Between DaVinci Resolve Pages

The order of sizing operations between DaVinci Resolve pages is a bit more nuanced. It's important to understand which sizing operations happen in the Fusion page, and which happen after, so you know which effects alter the image that's input to the Fusion page, and which effects alter the page's output.

For example, lens correction, while not strictly sizing, is nonetheless an effect that changes how the image begins in your Fusion composition. However, the Edit or Cut page stabilization function is an effect that comes after the Fusion page, so it does not appear in the composition you're creating.

The order of sizing effects in the different pages of DaVinci Resolve can be described as follows:



Sizing with Compound and Fusion Clips

Another way to modify the resolution before clips get handed off from the Edit page to the Fusion page is to create a compound clip or a Fusion clip. Both compound clips and Fusion clips change the working resolution of the individual clips to match the Timeline resolution. For instance, if two 4K clips are stacked one on top of the other in an HD timeline, creating a compound or Fusion clip resizes the clips to HD. The full resolution of the individual 4K clips is not available in Fusion and is therefore handed off to the Color page at the rescaled size. To maintain the full resolution of source clips, bring only one clip into the Fusion page from the Edit or Cut page Timeline, and then bring other clips into the Fusion composition using the Media Pool. Of course, if your clips are full HD and your timeline is full HD, then creating a Fusion clip or compound clip does not affect the resolution.



Fusion timeline sizing for Fusion and Compound clips

Color Bit Depths

The term bit depth describes how many colors are available in the color palette used to make up an image. The higher the bit depth, the greater the precision of color in the image, and therefore the greater the color reproduction. The higher precision is most apparent in gradients with subtle changes. Lower bit-depth gradients have noticeable banding artifacts, whereas higher bit-depth images can reproduce more colors, so fewer, if any, banding artifacts occur. The Fusion page within DaVinci Resolve always uses 32-bit float bits per channel precision to process images. However, in Fusion Studio you can choose to process images with 8-bit integer, 16-bit integer, 16-bit float, and 32-bit float bits per channel. Although always working at 16-bit float or 32-bit float will produce the best quality, it may be more efficient to use a lower bit depth if your images are 8-bit or 16-bit integer formats to begin with.

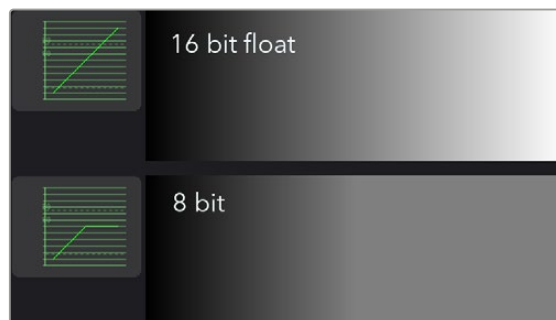
Understanding Integer vs. Float

Generally, 8-bit integer color processing is the lowest bit depth you'll come across for video formats. 8-bit images come from older or consumer-grade video equipment like mobile phones and camcorders. If you try to perform any significant gamma or color correction on 8-bit images, you can often see more visible banding.

16-bit integer color depth doubles the amount of precision, eliminating problems with banding. Although you can select 16-bit integer processing for an 8-bit clip, it does not reduce banding that already exists in the original file. Still, it can help when adding additional effects to the clip. This sounds like the best solution until you realize that many digital cameras like Blackmagic Design URSA Mini Pro

and others record in formats that can capture over-range values with shadow areas below 0.0 and super highlights above 1.0, which are truncated in 16-bit integer.

The 16-bit float color depth sacrifices a small amount of the precision from standard 16-bit integer color depth to allow storage of color values less than 0 and greater than 1.0. 16-bit float, sometimes called half-float, is most often found in the OpenEXR format and contains more than enough dynamic range for most film and HDR television purposes yet requires significantly less memory and processing time than is required for full float, 32-bit images.



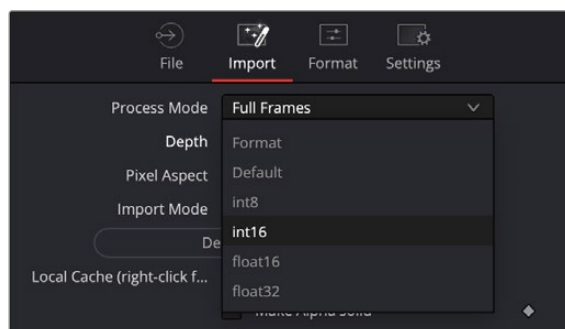
Preserving over-range values allows you to change exposure while maintaining highlights

Processing at 32-bit float can work with shadow areas below 0.0 and highlights above 1.0, similar to 16-bit float, except with a much greater range of precision but also much greater memory and processing requirements.

Setting Color Depth in Fusion Studio

As we said earlier, DaVinci Resolve always processes at 32-bit float bits per channel; however, you can use less memory and still achieve more-than-acceptable results using the Performance Mode setting located in the User > Playback Preferences panel.

Fusion Studio automatically uses the color depth that makes the most sense for each file format. For example, if you read in a JPEG file from disk, then the color depth for the Loader is set to 8 bits per channel. Since the JPEG format is an 8-bit format, loading the image at a greater color depth would generally be wasteful. If a 16-bit TIFF is loaded, the color depth is set to 16 bits. Loading a DPX file defaults to 32-bit float, whereas OpenEXR generally defaults to 16-bit float. However, you can override the automatic format color depth using the settings found in the Import tab of the Loader node's Inspector. The Loader's Inspector, as well as the Inspector for images generated in Fusion (i.e., text, gradients, fast noise, and others), has a Depth menu for 8-bit, 16-bit integer, 16-bit float, and 32-bit float.

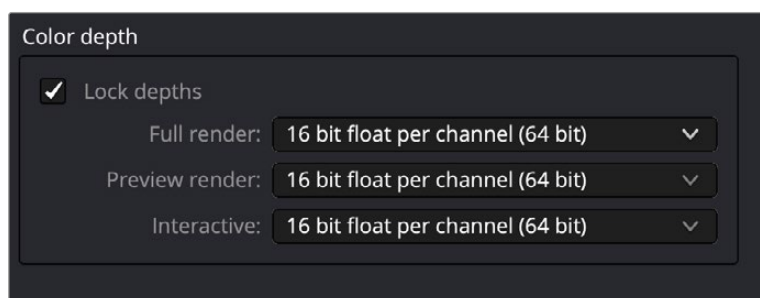


The Loader's Inspector Color Bit Depth settings

Configuring Default Color Depth Preferences

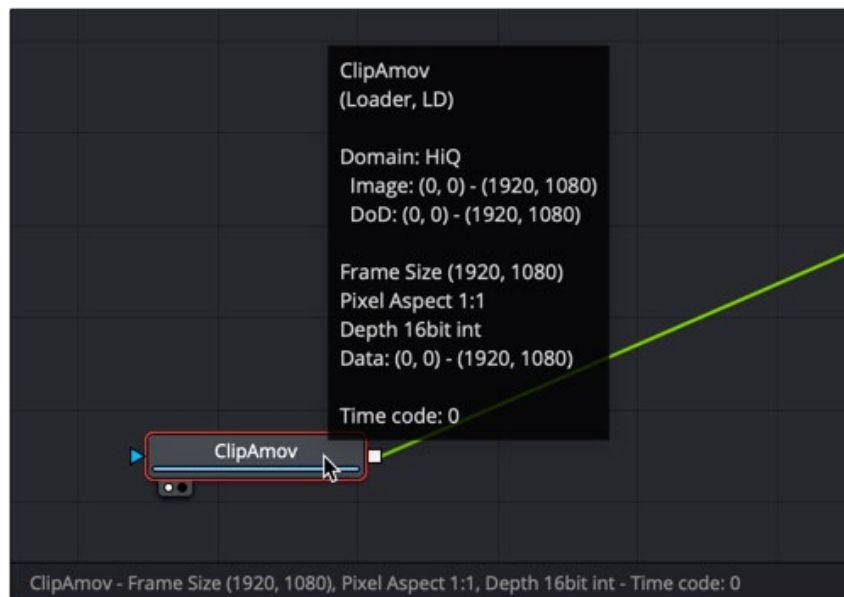
The default color depth setting forces the tool to process based on the settings configured in the Node Editor's Frame Format preferences. These are used to set a default value for color depth, applied when a Generator tool is added to the Node Editor. There are three drop-down menus to configure color depth in the preferences. They specify the different color depths for the interactive session, final renders, and preview renders.

To improve performance as you work on your comp, you can set the Interactive and Preview depth to 8-bits per channel, while final renders can be set to 16-bit integer. However, if your final render output is 16-bit float or 32-bit float, you should not use the integer options for the interactive setting. The final results may look significantly different from interactive previews set to integer options.



The Frame Format Color Depth settings

If you aren't sure what the color depth process is for a tool, you can position the pointer over the node's tile in the Node Editor, and a tooltip listing the color depth for that node will appear on the Status bar.



Hover over a node to view its Color Bit Depth setting.

TIP: When working with images that use 10-bit or 12-bit dynamic range or greater, like Blackmagic RAW or Cinema DNG files, set the Depth menu in the Inspector to 16-bit float or 32-bit float. This preserves highlight detail as you composite.

Combining Images with Different Color Depths

You can combine images with different color depths in a single composition. When images of different color depths are combined, the image from the background input of the node determines the bit depth output, and the foreground image is adjusted to match.

Advantages of Floating-Point Processing

There are two major advantages to floating-point processing that make the additional RAM requirements and longer render times worth your while. The first benefit is that floating-point values are more accurate than integer values. The second benefit is the preservation of shadow and highlight values that go beyond the normal tonal range.

Greater Accuracy

Using 16- or 32-bit floating-point processing prevents the loss of accuracy that can occur when using 8- or 16-bit integer processing. The main difference is that integer values cannot store fractional or decimal values, so rounding occurs in all image processing. Floating-point processing allows decimal or fractional values for each pixel, so it is not required to round off the values of the pixel to the closest integer. As a result, color precision remains virtually perfect, regardless of how many operations are applied to an image.

If you have an 8-bit pixel with a red value of 75 (dark red) and that pixel is halved using a Color Correction tool, the pixel's red value is now 37.5. Since you cannot store decimal or fractional values in integers, that value is rounded off to 37. Doubling the brightness of the pixel with another Color Correction tool should bring back the original pixel value of 75 but because of rounding 37×2 is 74. The red value lost a full point of precision due to integer rounding on a very simple example. This is a problem that can result in visible banding over several color corrections. Similar problems arise when merging images or transforming them. The more operations that are applied to an image, the more color precision is lost to rounding when using 8- or 16-bit integer processing.

Accessing Extended Highlights and Shadows

Increasingly more productions are capturing out-of-range images thanks to digital cinema cameras like the Blackmagic URSA Mini Pro and even the Pocket Cinema 6K camera. These cameras capture very high dynamic range RAW images and maintain color detail even in heavily over or underexposed frames. The extended white color detail can also give very nice, natural results when blurred, glowed, color corrected, or even just when faded or dissolved. While it is possible to work with these RAW images using integer data, doing so results in the loss of the extended range values, losing all detail in the highlights and shadows. Float processing makes working with logarithmic RAW images considerably easier by preserving highlight and shadow detail.

If you have an 8-bit pixel that has a red value of 200 (bright red) and a Color Gain tool is used to double the brightness of the red channel, the result is 200×2 , or 400. However, 8-bit color values are limited to a range of 0 through 255. So the pixel's value is clipped to 255, or pure red. If now the brightness is halved, the result is half of 255, or 127 (rounded), instead of the original value of 200.

When processing floating-point colors, pixel values brighter than white or darker than black are maintained. There is no value clipping. The pixel is still shown in the viewer as pure red, but if float processing is used instead of 8-bit, the second operation where the gain was halved would have restored the pixel to its original value of 200.

Using Float with 8-Bit HD Video

There is also some value to using float color depths with an 8-bit HD video when the images require a lot of color correction. Using float helps maintain precision by avoiding the rounding errors common to 8-bit processing, as described above.

Detecting Extended Highlight and Shadow Values

Although floating-point processing preserves extended values below 0.0 and greater than 1.0, also called “out-of-range values,” the viewer still displays them as black or white. This can make it difficult for you to determine the overall dynamic range of an image.

To discover whether there are out-of-range values in a viewed image:

Right-click in the viewer and choose Options > Normalized Color Range.



Use the Normalized Color Range pop-up menu to detect out-of-range images.

Enabling this display mode rescales the color values in the image so that the brightest color in the image is remapped to a value of 1.0 (white), and the darkest is remapped to 0.0 (black).

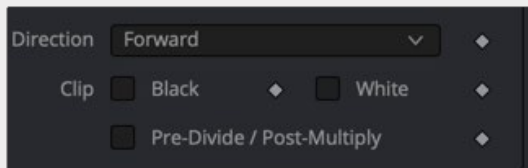
The 3D Histogram subview can also help visualize out-of-range colors in an image.

For more information, see *Chapter 69, “Using Viewers.”* in the DaVinci Resolve Reference Manual, or Chapter 7 in the Fusion Reference Manual.

Clipping Out-of-Range Values

When processing in floating point, there may be situations where the out-of-range values in an image need to be clipped. The Brightness/Contrast tool provides checkboxes that can be used to clip out-of-range values to 0 or 1.

For example, there may be files that contain out-of-range alpha values. Since the alpha channel represents the opacity of a pixel, it makes little sense to be more than completely transparent or more than fully opaque, and compositing such an image may lead to unexpected results. To easily clip alpha values below 0 and above 1, add a Brightness/Contrast toolset to Clip Black and Clip White, with only the Alpha checkbox selected.



Clip White and Clip Black settings in Brightness/Contrast can be used to clip mattes.

Alternatively, you can clip the range by adding a Change Depth node and switching to 8-bit or 16-bit integer color depths.